frontiers

Supplementary Material

1 SUPPLEMENTARY TABLES AND FIGURES

1.1 Tables

Table S1. Pre-optimization pinned-knee flexor torque-prediction goodness of fit per test, comparing the three pre-optimization torque predictions of Figure 7: the Hunt-equation rigid-body prediction, the unoptimized prediction of the improved BPA characterization, and the back-calculated (hybrid) torque. RMSE and maximum absolute residual (Max.) are in N m, and FVU denotes the fraction of variance unexplained.

		Hunt equation			**Improved BPA**			**Back-calculated (hybrid)**		
l_0 (cm)	l_{ten} (mm)	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
48.5	–	5.779	1.1761	9.798	4.871	0.8354	9.865	1.635	0.0991	3.842
41.0	–	6.633	3.2313	11.409	6.074	2.7090	9.262	2.187	0.1184	4.243
45.7	–	5.315	0.9408	9.991	4.465	0.6640	9.536	1.246	0.0409	2.275
40.6	35	3.945	0.6750	7.527	3.226	0.4512	7.376	2.085	0.0755	3.073

Table S2. Muscle origin and insertion locations for different models. Origin is in femur reference frame, insertion is in the tibia reference frame.

Muscle	Origin/Insertion	Model	X (m)	Y (m)	Z (m)
Flexor	origin	Pinned knee	-0.075	0.100	0.0328
Flexor	insertion	Pinned knee	-0.05011	-0.045	0.0326
Extensor	origin	Pinned knee	0.030	-0.050	0
Extensor	insertion	Pinned knee	0.0425	-0.076	0
Bifemsh	origin	Gait2392	0.005	-0.211	0.023
Bifemsh	insertion	Gait2392	-0.03	-0.036	0.029
Vastus int.	origin	Gait2392	0.029	-0.192	0.031
Vastus int.	insertion	Gait2392	moving	moving	0.0018
Flexor, 10 mm	origin	Biomimetic knee	-0.050	-0.045	0.0328
Flexor, 10 mm	insertion	Biomimetic knee	-0.0279	-0.046	0.0328
Extensor, 10 mm	origin	Biomimetic knee	0.040	0.035	0
Extensor, 10 mm	insertion	Biomimetic knee	0.021	-0.072	0
Flexor, 20 mm	origin	Biomimetic knee	-0.050	-0.045	0.0328
Flexor, 20 mm	insertion	Biomimetic knee	-0.0279	-0.046	0.0328

1.2 BPA routing and via points

The tables in this section list the routing via points of the BPA configurations reported in the main text. All coordinates are coded in meters in the project's MATLAB sources and are reported here in millimeters, rounded to 0.01 mm. Knee angle $\theta_{\mathbf{k}}$ follows the model convention with flexion negative. Femur-side points (path rows before the crossing index) are specified in the femur-referenced instantaneous-center-of-rotation (ICR) frame and are used untransformed at every knee angle, while tibia-side points are specified in the tibia θ_1 frame and are mapped into the ICR frame at each pose with the measured ICR migration, fitted as cubic splines of the tibia-to-ICR offsets over the knee range. The pinned-knee bracket reference points use the knee-ICR and hip frames noted per row, and the pinned-knee route endpoints (origins and insertions) are listed in Table S2.

Bracket reference points of the pinned-knee BPA routes, in mm. The pinned flexor route runs from origin to insertion with a compliant bracket at each end, and the pinned extensor route additionally wraps on two fixed arcs ($R_1 = 12\,\mathrm{mm}$, active for $-23^\circ \leqslant \theta_{\mathbf{k}} \leqslant 27^\circ$, and $R_2 = 40\,\mathrm{mm}$, active for $\theta_{\mathbf{k}} \leqslant -23^\circ$).

Table S3. Pinned-knee bracket reference points in millimeters.

Configuration	Point	Frame	x (mm)	y (mm)	z (mm)
Pinned flexor	$\mathbf{p}_{bri}$, insertion bracket	knee ICR	-48.11	-107.81	13.80
Pinned flexor	$\mathbf{p}_{br2}$, origin-side second bracket	hip	-52.61	0.00	75.06
Pinned extensor	$\mathbf{p}_{br}$, rib midpoint, medial side	hip	-3.84	-46.44	62.50

Routing via points of the biomimetic-knee $10\,\mathrm{mm}$ BPA routes, in mm. The flexor route is a two-point path (crossing index 2), and the extensor route is the eight-point path of the extensor test (crossing index 5, nominal resting length $520\,\mathrm{mm}$) whose femur- and tibia-side contacts engage progressively with flexion. The active-range column lists the knee angles over which each point participates in the path.

Table S4. Biomimetic-knee $10\,\mathrm{mm}$ routing points in millimeters. Active ranges identify when each contact participates.

Point	Description	Frame	x (mm)	y (mm)	z (mm)	Active $\theta_{\mathbf{k}}$
Flexor, $10\,\mathrm{mm}$ **(two-point route, resting length** $415\,\mathrm{mm}$**)**						
p_1	origin	femur	-50.00	35.00	50.00	all $\theta_{\mathbf{k}}$
p_2	insertion	tibia θ_1	-12.24	-8.87	27.87	all $\theta_{\mathbf{k}}$
Extensor, $10\,\mathrm{mm}$ **(eight-point route, resting length** $520\,\mathrm{mm}$**)**						
p_1	origin	femur	40.00	35.00	0.00	all $\theta_{\mathbf{k}}$
p_2	mounting-base contact	femur	75.90	-274.76	0.00	all $\theta_{\mathbf{k}}$
p_3	femur channel contact	femur	59.82	-374.27	0.00	$\theta_{\mathbf{k}} \leqslant -9.19^\circ$
p_4	femoral condyle contact	femur	39.55	-421.83	0.00	$\theta_{\mathbf{k}} \leqslant -69.8^\circ$
p_5	tibia contact	tibia θ_1	58.71	25.00	0.00	$\theta_{\mathbf{k}} \leqslant -9.19^\circ$
p_6	tibia contact	tibia θ_1	58.71	12.28	0.00	all $\theta_{\mathbf{k}}$
p_7	tibia tendon contact	tibia θ_1	54.00	-6.12	0.00	all $\theta_{\mathbf{k}}$
p_8	patellar ligament ring	tibia θ_1	36.04	-28.44	0.00	all $\theta_{\mathbf{k}}$

Routing via points of the $20\,\mathrm{mm}$ route redesigns, in mm. The flexor design uses a staggered pair of BPA paths: the second BPA keeps the z-mirrored distal endpoint while both origins sit off the same side of the knee, and each route carries one tibial wrap point placed at the BPA centerline standoff from the tibial surface while contact is active. The extensor design is the clip-on-cam route; the liftoff column lists the knee angle above which each contact leaves the path as the knee extends, with all contacts engaged at maximum flexion.

Table S5. Routing points of the $20\,\mathrm{mm}$ redesigns in millimeters. Liftoff is the angle above which a contact leaves the route during extension.

Point	Description	Frame	x (mm)	y (mm)	z (mm)	Liftoff $\theta_{\mathbf{k}}$
Flexor, $20\,\mathrm{mm}$ **(two-BPA mirrored route, two BPAs, resting length** $561.49\,\mathrm{mm}$**, tendon** $25.00\,\mathrm{mm}$**)**						
p_1	origin, BPA 1	femur	-125.00	135.00	72.08	end point
p_{end}	insertion, BPA 1	tibia θ_1	-18.72	-32.69	38.96	end point
$p_1^{(2)}$	origin, BPA 2	femur	-125.00	135.00	-5.85	end point
$p_{end}^{(2)}$	insertion, BPA 2	tibia θ_1	-18.72	-32.69	-38.96	end point
Extensor, $20\,\mathrm{mm}$ **(clip-on-cam route, two BPAs, resting length** $827.74\,\mathrm{mm}$**, tendon** $92.49\,\mathrm{mm}$**)**						
p_1	origin (fixed)	femur	40.00	137.00	0.00	end point
p_2	mounting-base contact	femur	83.00	-271.70	0.00	-21.52°
p_3	femur contact	femur	66.60	-427.80	0.00	-54.34°
p_4	femur contact	femur	54.70	-443.60	0.00	-89.80°
p_5	femoral condyle contact	femur	38.80	-454.40	0.00	-110.81°
p_6	cam contact (optimized)	tibia θ_1	116.32	15.04	0.00	-18.89°
p_7	cam contact (optimized)	tibia θ_1	131.32	-18.97	0.00	none
p_8	cam contact (optimized)	tibia θ_1	127.04	-47.87	0.00	none
p_9	distal ring (insertion)	tibia θ_1	34.42	-197.10	0.00	end point

1.3 Figures

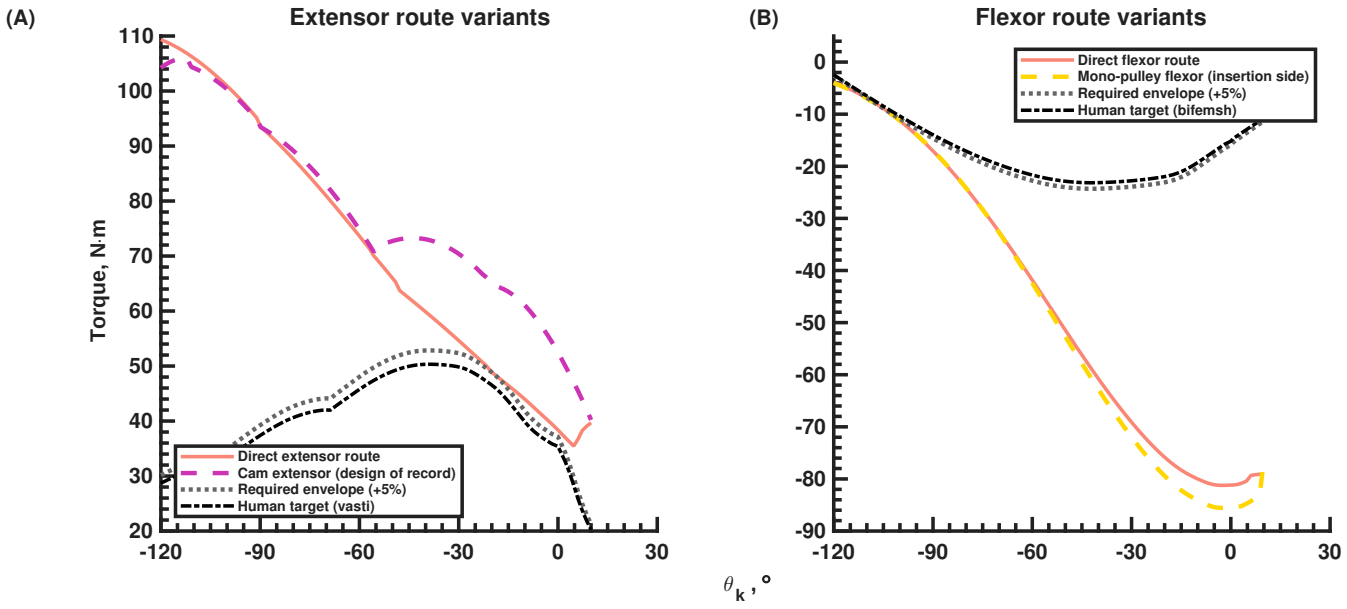


Figure S1: Predicted redesigned extensor (A) and flexor (B) torque against the human reference and required $5\,\%$ design margin.

1.4 Contact and Release Angles of the Wrapped Route

The route-elimination rule that schedules the wrapped extensor route is a route-topology state machine rather than a shortest-path solver: the fully flexed route carries every candidate contact, and each contact is released at the first more-extended knee angle at which the straight chord between its surviving neighbors can replace it without entering the exclusion envelopes. This section states the calculation in generic terms, so that it applies to any wrapped actuator route on a rotating joint pair; the extensor route of Table S5 gives the cam-route example; its release schedule follows the same rule on the cam geometry.

The route is stored as a fixed number of rows $p_1, \ldots, p_9$: rows p_1 through p_5 are femur-frame points, rows p_6 through p_9 are tibia-frame points, and the segment from p_5 to p_6 crosses from one body frame into the other. The endpoints p_1 and p_9 are the actuator origin and insertion, and they are always active. Each interior row is either active, a genuine wrap contact, or eliminated; an eliminated femur-side row repeats the preceding active femur row, and an eliminated tibia-side row repeats the neighboring active tibia row, so the route keeps a fixed size and its segment chain crosses the joint exactly once. Let $\mathcal{A}(\theta_\mathbf{k}) \subseteq \{2, \ldots, 8\}$ denote the active interior rows at knee angle $\theta_\mathbf{k}$. The sweep over the design grid, here 100 poses from -120° to 10° at a spacing of 1.31°, proceeds from full flexion toward extension, each pose inherits the active set of the previous pose, and rows are only ever removed, so $\mathcal{A}$ shrinks monotonically as the knee extends.

At every pose, every active interior row is tested between its nearest active anchors,

$$i_j^- = \max\{i < j : i \in \mathcal{A} \cup \{1\}\}, \qquad i_j^+ = \min\{i > j : i \in \mathcal{A} \cup \{9\}\}, \tag{S1}$$

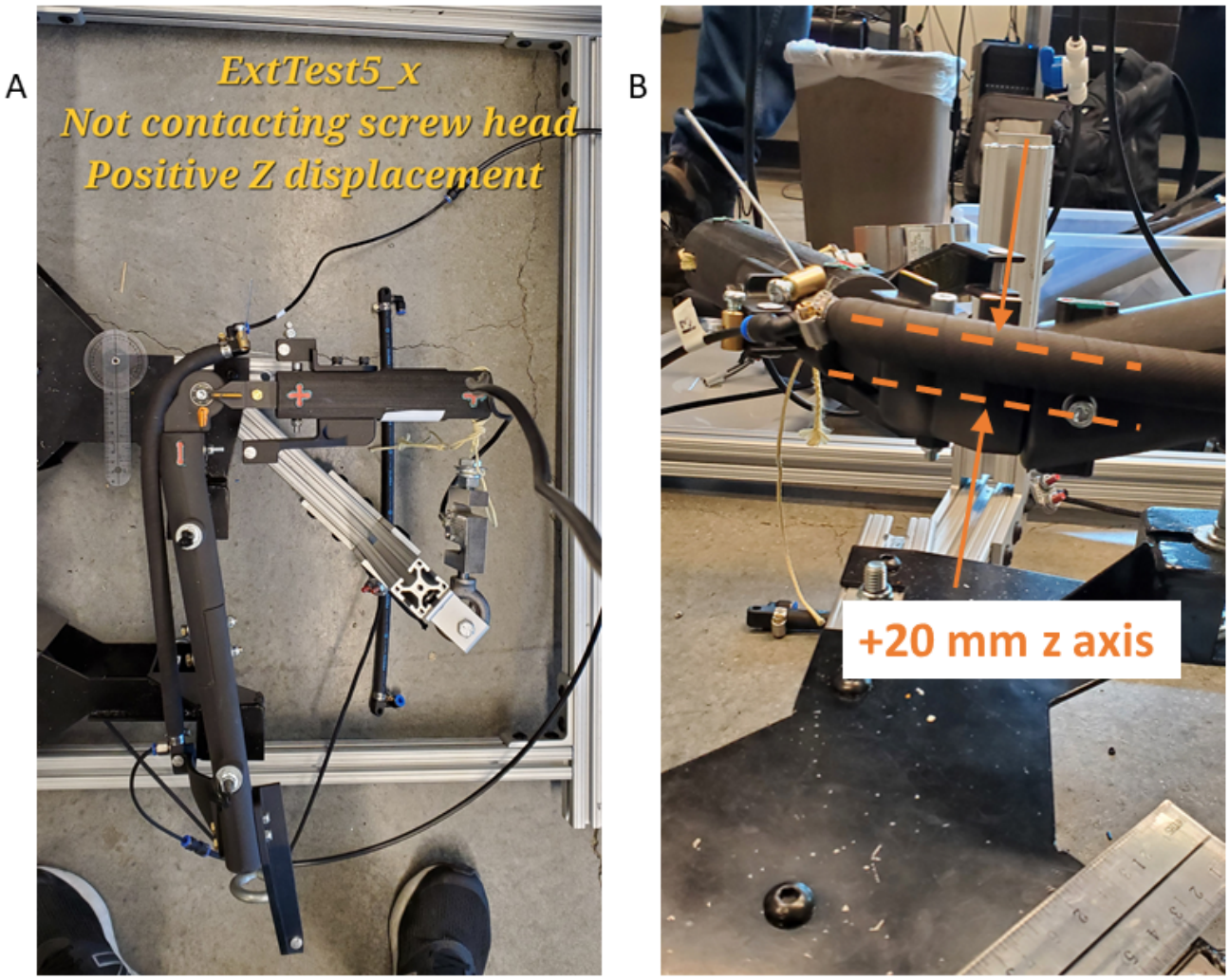


Figure S2: Pinned-extensor actuator slippage: top view (A) and side view (B). The lateral displacement changes the force path and moment arm.

with the anchors expressed in the candidate's native frame through the per-angle joint transforms. Write A, B, and C for the planar, sagittal positions of $p_{i_j^-}$, $p_{i_j^+}$, and p_j, and let $s(x, y)$ denote the signed angle from vector x to vector y, positive counterclockwise, evaluated as a single $\mathrm{atan2}$ of the planar cross product and the dot product so that no principal-value seam can corrupt the comparison. The elimination margin of row j is

$$\mu_j(\theta_{\mathbf{k}}) = \begin{cases} s(C - A,\ B - A)\,, & j \leqslant 5, \\ s(A - B,\ C - B)\,, & j \geqslant 6, \end{cases} \tag{S2}$$

a signed angle between the through-contact leg and the bypass chord, measured in the native wrap sense of the candidate's side of the crossing segment; the two cases differ only in that sense of rotation, because the femur-side contacts turn the route one way around the distal femur and the tibia-side contacts turn it the opposite way around the proximal tibia. The rotation rule releases the contact once the margin turns

positive, the orientation at which the chord, rather than the contact, provides the outer boundary of the route on that side. The gate is opened slightly early,

$$\mu_j(\theta_{\mathbf{k}}) > -\Delta, \qquad \Delta = 1.5^{\circ}, \tag{S3}$$

as a release hysteresis: in the last poses before the contact becomes collinear with its chord, the computed moment arm of the wrapping segment collapses, and releasing the contact slightly before that orientation, with the collision gate below still controlling validity, removes the artifact without changing the route family.

Opening the rotation gate is necessary but not sufficient. A removal additionally requires the straight bypass chord $\overline{AB}$ to clear the exclusion envelopes of the candidate's side,

$$\operatorname{dist}\big(\overline{AB}_{\mathrm{trim}},\, \mathcal{E}_j\big) \;\geqslant\; g, \qquad g = 0.5\,\mathrm{mm}, \tag{S4}$$

where dist is the planar clearance between the trimmed chord and the envelope and tangency is allowed. The chord is trimmed by $2\,\mathrm{mm}$ at each end, capped at $45\,\%$ of its length, because a legitimate contact departs from the clearance boundary itself. On the femur side the envelopes $\mathcal{E}_j$ are the femoral-cylinder clearance circle, the condyle clearance polygon, and the femoral-wall span; on the tibia side they are the two tibial-cylinder clearance circles and the wall band between them. Every envelope is the normal offset of a bone profile by the fully inflated BPA radius R_{BPA},

$$\mathbf{q}(\vartheta) = \mathbf{e}(\vartheta) + R_{\mathrm{BPA}}\,\hat{\mathbf{n}}(\vartheta), \tag{S5}$$

where $\mathbf{e}(\vartheta)$ traces the physical profile and $\hat{\mathbf{n}}(\vartheta)$ is its outward normal; the offset curve shares the tangent of the physical profile, its radius of curvature is $\rho(\vartheta) = \rho_{\mathrm{phys}}(\vartheta) + R_{\mathrm{BPA}}$, and the candidate contact positions themselves are seeded at tangent points of this curve. Because the envelopes are drawn at the fully inflated radius while the routed BPA is contracted at the release angles, each envelope radius may be relaxed by $\rho = 3\,\mathrm{mm}$ inside the gate, floored at the bone profile plus $0.5\,\mathrm{mm}$; a rotation-gate opening that still fails the relaxed collision gate indicates that the seed placement, not the gate, must move.

At the first pose of the sweep, full flexion, the seed route is settled to a fixpoint: every contact whose two gates already open there is removed, and removals repeat until no further gate opens, so that the sweep starts from a valid topology; a colinearity guard may additionally remove a seed contact that starts the sweep already aligned with its neighbors, and its tolerance is zero in the design of record, so every seed row of Table S5 starts active. At every later pose at most one contact is released: if several rows open both gates in the same pose, the row with the largest elimination margin, the one whose rotation rule is most decisively open, is released first, and the remaining rows are retested at the next pose. The release angle of row j is recorded as the first grid pose at which the row is inactive. The activation angle is read from the same record with the sweep reversed, as the last pose at which the row is active, which is the pose at which the contact re-engages when the route is driven from extension toward flexion.

The same active-set record drives the geometric bend measure that scales the fictive-strain wrapping loss of Equations (25) and (27) inside the stiffness-aware force balance. Each wrap family, the femoral-cylinder contact, the femoral-condyle curve, the tibial contact curve, and the crossing-point curve at the ICR, contributes an arc length over its effective radius, and the total

$$b(\theta_{\mathbf{k}}) = \sum_{f} R_f(\theta_{\mathbf{k}})\,\alpha_f(\theta_{\mathbf{k}}) \tag{S6}$$

is the bend signal handed to the force model. Each family's turn angle α_f ramps with knee angle between the two route-state events that bound it, holding zero before the first event and its plateau after the second, so the topology state and the bending signal are two readings of one record. The formulation is joint-agnostic: it requires only a fixed route representation with one crossing segment, per-angle transforms between the two body frames, and clearance envelopes on each side, so it carries unchanged to the other wrapped routes of this program.

1.5 Configuration-Survey Coordinates and Actuator Sizes

Table S6. Graded coordinates and nominal BPA sizes for the 27-site survey. At nominal 40 mm sites, the baseline uses four parallel 20 mm BPAs.

Site	Primary DoF	Secondary DoF	Dia. (mm)	Baseline (mm)
ercspn_r	lumbar ext.	lumbar bend.	40	4×20
intobl_r	lumbar ext.	lumbar bend.	20	1×20
extobl_r	lumbar ext.	lumbar bend.	20	1×20
bifemlh_r	knee	hip flex.	20	1×20
semimem_r	knee	hip flex.	40	4×20
sar_r	knee	hip flex.	10	1×10
tfl_r	knee	hip flex.	10	1×10
rect_fem_r	knee	hip flex.	20	1×20
grac_r	knee	hip flex.	10	1×10
bifemsh_r	knee	hip flex.	20	1×20
vas_int_r	knee	hip flex.	40	4×20
soleus_r	ankle	knee	40	4×20
med_gas_r	ankle	knee	40	4×20
lat_gas_r	ankle	knee	20	1×20
tib_post_r	subtalar	ankle	40	4×20
tib_ant_r	ankle	subtalar	20	1×20
per_brev_r	subtalar	ankle	10	1×10
per_long_r	subtalar	ankle	20	1×20
per_tert_r	subtalar	ankle	10	1×10
flex_dig_r	MTP	ankle	10	1×10
flex_hal_r	MTP	ankle	10	1×10
ext_dig_r	MTP	ankle	10	1×10
ext_hal_r	MTP	ankle	10	1×10
glut_max_r	hip flex.	hip add.	40	4×20
add_mag_r	hip add.	hip flex.	40	4×20
iliacus_r	hip flex.	hip add.	20	1×20
psoas_r	hip flex.	hip add.	20	1×20

1.6 Cam-Extensor Routing Geometry

The route is shown throughout knee extension in Figures S3 and S4. Gray anatomical silhouettes locate the femur and tibia; colored outlines show the routing-clearance envelopes.

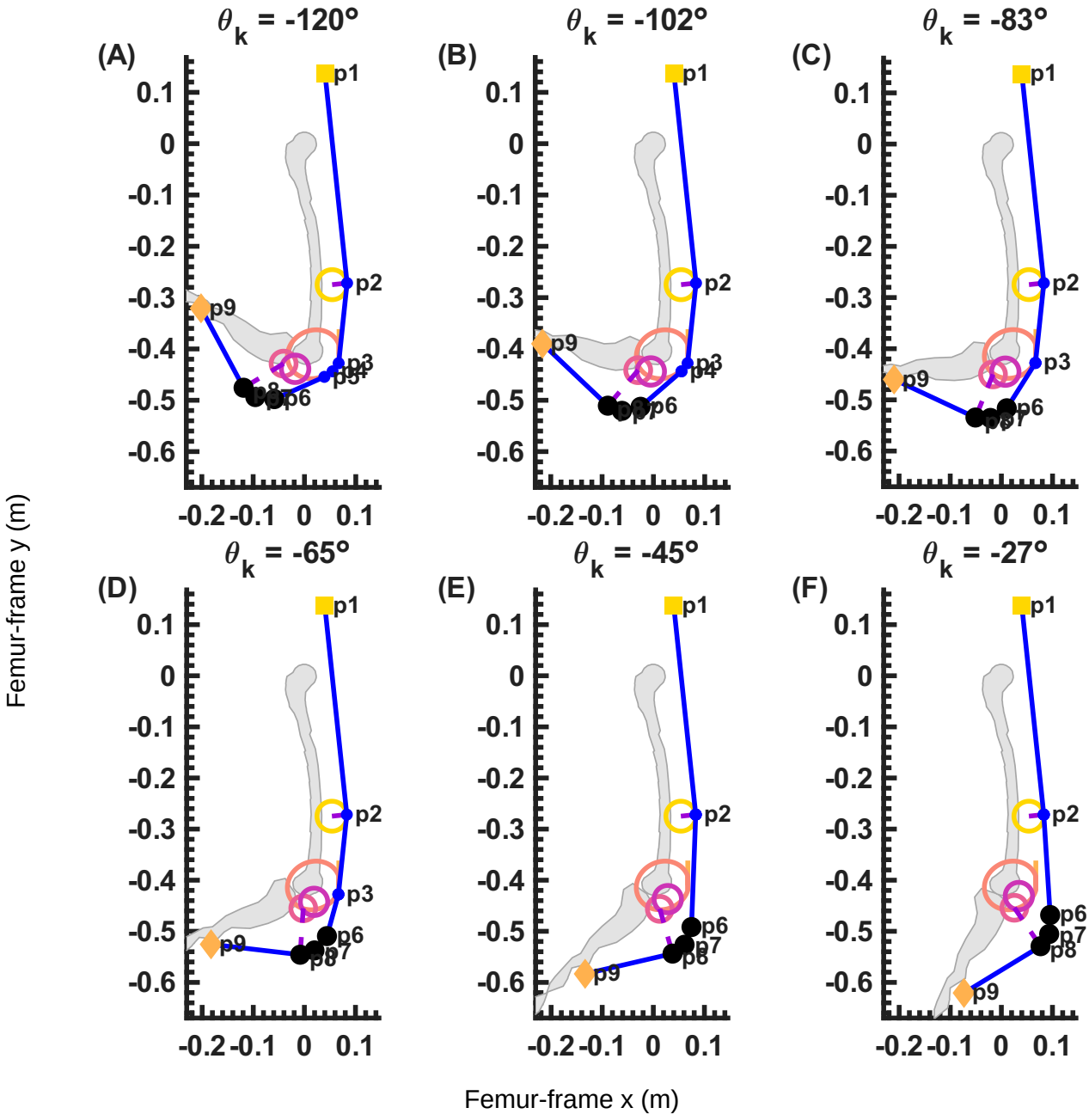
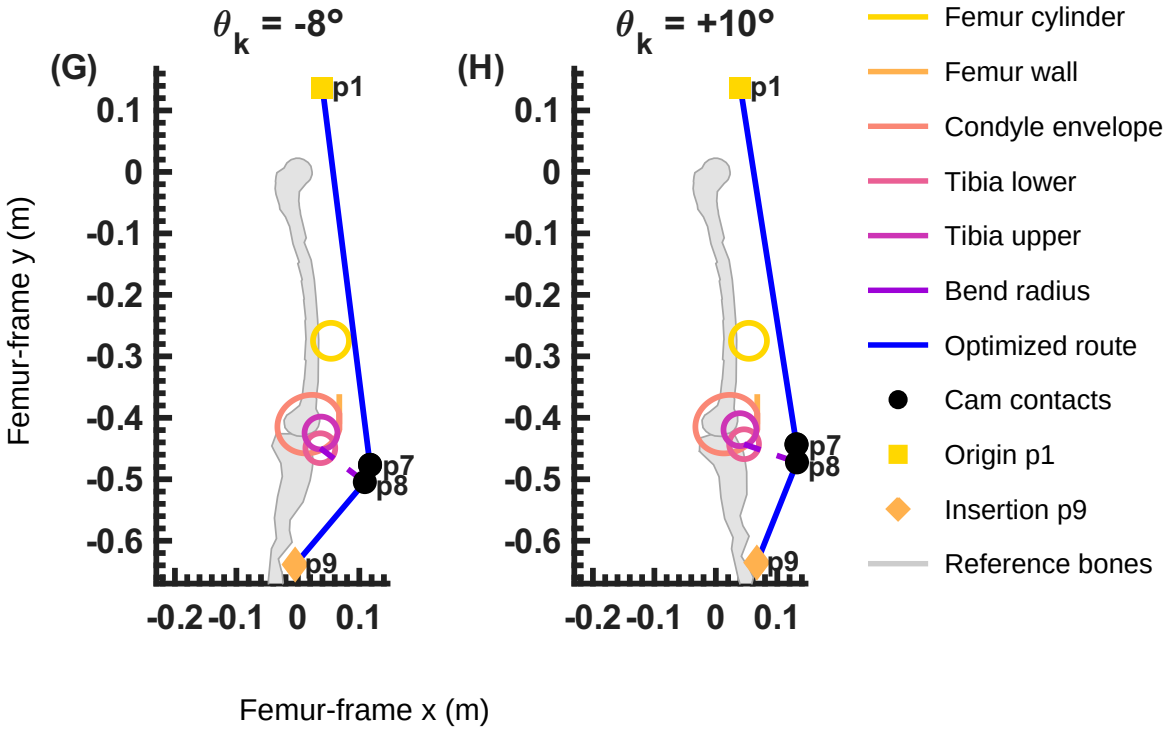


Figure S3: Cam-extensor route at six knee poses, A–F, from $-120°$ to $-27°$.



Cam-drum radius: 96.9 mm

Release angles: p5 -110.8 deg; p4 -89.8 deg; p3 -54.3 deg;
p2 -21.5 deg; p6 -18.9 deg. Gray bones are anatomical references.

Figure S4: Cam-extensor route at -8° (G) and 10° (H), with the legend and contact-release schedule.